Smart Hostel Room Allocation System

Project Proposal submitted for UCS503

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1 Introduction / Problem Statement

- Our institute already runs FretBox for hostel administration: it holds student master data and handles fees, complaints and gate passes.
- FretBox does not, however, let students choose where they live. Rooms are assigned centrally, and friend groups coordinate their requests informally over WhatsApp, so that context is lost by the time it reaches the warden.
- Students have no visibility into room or cluster availability until check-in.
- Multiple groups wanting the same rooms at the same time creates a risk of double-booking, and room payment is tracked separately from allocation, so a "reserved" room is not always a confirmed one.
- These issues only grow as enrollment and hostel occupancy increase every year.
- This project proposes a self-service web platform, layered on FretBox data, where students form groups, submit preferences, and are allocated a room automatically, with payment required before final confirmation. All student and hostel records are drawn from FretBox; nothing is entered or maintained by hand.

2 Objectives

Our main goal is to design and deploy a web-based hostel room allocation system that adds transparent, student-driven allocation on top of the institute's existing FretBox deployment. Specific objectives:

- Replace centrally-assigned allocation with a self-service digital process, driven by data already held in FretBox.
- Ensure fair, conflict-free room allocation, even when many groups apply at once.
- Tie payment directly to booking confirmation, so a booking is only final once it is paid for.
- Give wardens better visibility and control over the allocation process.
- Deliver a working, deployable system within the project timeline.

3 Methodology / Scope of Work

The project will be delivered as a web application with the following scope:

- Student registration and login, with profiles sourced from FretBox rather than manual entry.
- Group formation and joining via a shareable code (up to four members per group).
- A visual, floor-wise room/cluster map built from imported hostel data, showing live occupancy status.
- Ranked preference submission by the group leader (3–5 preferred clusters).
- An automated allocation engine that matches groups to rooms based on group size and preference order.

- Conflict handling so that concurrent requests for the same room are resolved safely, without double allocation.
- A payment step, integrated with a payment gateway, required from every group member before a booking is confirmed.
- An admin/warden dashboard to view all allocations and payments, with the ability to manually reassign a room if required.
- A scheduled FretBox data sync, plus packaging so the system can be set up and run with minimal configuration.

Testing and Validation

- Confirming no two groups can be allocated the same room, even under simultaneous requests.
- Checking that the full flow, from group formation to confirmed and paid booking, works end-to-end.
- Verifying that admin overrides update room status correctly.
- Running a live demo with a small set of test groups, including a case where two groups compete for the same room.

Key Risks and Mitigations

- Incomplete group payment — handled by releasing an unpaid reservation back to the pool after a time limit.
- Low student adoption — addressed by keeping the interface simple and familiar.
- Payment gateway unavailability during a demo — addressed with a fallback demo mode.

4 Timeline

Planned over twelve weeks, with one buffer week for testing and fixes:

Week	Task	Deliverable
1	Requirements, planning, and system/UI design	Approved plan and wireframes
2	FretBox data study; import and sync design	Data mapping and sync specification
3	Student registration, login, and profile module	Working auth and profile module
4	Group formation and joining via group code	Working group module

Week	Task	Deliverable
5	Room/cluster availability map from imported data	Live occupancy view
6	Allocation logic for 2- and 4-member groups	Core allocation engine
7	Allocation logic for 3-member groups and single applicants	Complete allocation engine
8	Redis locking and safeguards against double-booking	Conflict-safe booking flow
9	Payment gateway integration and booking confirmation	End-to-end paid booking flow
10	Admin/warden dashboard, overrides, and reconciliation	Admin module
11	Buffer week: integration testing, load testing, bug fixes	Test report
12	Documentation, deployment packaging, and final demo	Final system and report

5 Resources / Budget

- Team of three, with work split across frontend, backend/allocation logic, and payments/admin/testing.
- Standard laptops are sufficient; no specialised lab equipment or paid software is required.
- All tools planned for use are free or offer a free tier for development and demo purposes.
- No external funding required; the project runs entirely on personal or university-provided infrastructure.